

Saumya Shah

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Education

The University of Texas at Dallas

Dallas, Texas

January 2023 – December 2024

- Master of Science in Electrical Engineering
- **Relevant Coursework:** VLSI Design, ASIC Design, Physical Design Automation, STA, Functional Verification, Testing & Testable Design, Computer Architecture

Nirma University

Ahmedabad, India

July 2018 – June 2022

- Bachelor of Engineering in Electronics and Communication Engineering.

Technical Skills

- **Languages:** C++, C, Python
- **Tools & Technologies:** STM32CubeIDE, PlatformIO, KiCad, Arduino IDE, Git, Keil
- **Concepts:** Embedded Firmware Development, ARM Cortex-M Microcontrollers, Event-Driven Firmware Architecture, Real-Time Data Acquisition, PCB Manufacturing Automation, Gerber File Processing, PCB Panelization
- **Protocols:** I2C, SPI, UART, 1-Wire, Wi-Fi

Work Experience

Development Engineer

Express Fab USA

December 2025 – Present

- Developed a Python automation tool that programmatically adds tooling rails to customer-supplied Gerber files, bringing incoming PCB designs into compliance with panelization and pick-and-place (PnP) assembly requirements across a wide range of board geometries.
- Engineered adaptive rail profiles with curved trapezoidal edges and dynamic neck sizing to handle narrow-board edge cases while supporting both mouse-bite and V-score breakaway tab conventions.
- Generated machine-readable barcodes rendered directly as native board geometry on the rail silkscreen layer, enabling automated panel identification and end-to-end assembly traceability.
- Built robust file-parsing and file-type classification logic with standardized high-resolution millimeter output, ensuring reliable processing of fabrication, assembly, and position/bill-of-materials (BOM) files exported from multiple CAD tools.

Embedded Engineer

Winwinlabs

April 2025 – January 2026

- Built a real-time data acquisition and communication system on the ESP32-S3, using careful timing analysis and resource optimization to ensure consistent performance across deployments.
- Implemented automated OTA updates, configuration handling, and debug tooling, improving firmware maintainability and streamlining development workflows.
- Designed and brought up a custom PCB for the product, handling schematic capture, layout considerations, and hardware validation in line with embedded design best practices.

Research and Development Intern

Thruster, India

May 2021 – July 2021

- Designed a circuit with STM32F4 (Arm Cortex-M4) to drive a motor controller and capture coil current/voltage data for later analysis.
- Developed tailored algorithms to refine the SRM motor driver assessment process, cutting evaluation time by 30% and improving decision-making reliability.

Projects

Sobel Edge-Detection for Reconfigurable Computing

September 2024 – December 2024

- Implemented Sobel filter accelerator with HLS, optimized using synthesis directives.
- Integrated with ARM HPS for hardware/software co-design, reducing latency by 14%.
- Verified cycle-accurate timing and resource utilization with **QoR reports**.

Analysis of Branch Predictors and Cache Associativity

September 2023 – December 2023

- Conducted a comprehensive analysis of diverse cache design strategies, assessing the influence on x86 CPU performance. Subsequently, derived an optimal configuration yielding the lowest Cycle Per Instruction (CPI) while maintaining cost efficiency.
- Evaluated the impact of changing variables on branch predictors through experimentation using advanced features from GEM5.
- Generated python scripts to improve the benchmarks' execution speed by 30%.
- Explored cache coherence and branch prediction performance in GEM5, aligned with CPU fabric-level microarchitecture concepts.

Custom PCB Design for Embedded and Robotic Applications

October 2018 – December 2021

- Designed development boards featuring the LPC17xx microcontroller for embedded system development.
- Developed an opto-isolated, 4-channel MOSFET switching circuit for high-speed, reliable I/O control.
- Created custom PCB solutions integrating the STM32F4 Discovery board for advanced robotic applications.